

PahadiCart (पहाड़ीकार्टे)

Himachal Pradesh Mountain Hyperlocal Delivery Architecture:
HTML5 Live GPS Auto-Detection, Open-Meteo Weather Sync &
Universal Google Maps Deep Links.

TARGET MOUNTAIN REGIONS

Solan, Shimla Ridge, Dharamshala

OPEN SOURCE TECH STACK

**HTML5 GPS, Open-Meteo, Leaflet,
PWA**

LIVE PRODUCTION URL

<https://pahadicart.vercel.app>

GITHUB REPOSITORY

HACK-X01 / pahadicart (main)

1. Executive Summary & 5-Portal Architecture

PahadiCart is an indigenous mountain-first hyperlocal commerce platform engineered specifically for the steep ridges, winding road curves, and ancient staircase footpaths of Himachal Pradesh. Unlike conventional metropolitan delivery apps that rely on flat aerial distance (which miscalculates mountain travel times by over 200%), PahadiCart computes real mountain road curves and foot elevation climbs.

Portal Name	Live Production URL	Primary Stakeholder	Core Technical Capabilities
Super Hub Launcher	https://pahadicart.vercel.app/	Founders & Admins	Ecosystem switcher & 1-Click 5-Stage Lifecycle Order Simulator.
Super Admin Tower	https://pahadicart.vercel.app/admin/	Operations Managers	God View dispatch map, live Open-Meteo weather sync, triad order audit.
Customer PWA App	https://pahadicart.vercel.app/customer/	Consumers & Tourists	Compulsory GPS prompt, live mountain radar, staircase address notes.
Merchant Shop Portal	https://pahadicart.vercel.app/merchant/	Kirana Dukandars & Cooks	"Ghar Ki Ghanti" audio chime, automated voice call, 58mm thermal KOT.
Rider Mobile Cockpit	https://pahadicart.vercel.app/rider/	Scooter & Pedestrian Fleet	Turn-by-turn Google Maps navigation, batched route energy optimizer.

2. Compulsory HTML5 Live GPS Auto-Detection Flow

To ensure accurate staircase footpath routing and prevent deliveries to wrong mountain ridges, PahadiCart enforces a **Compulsory Location Access Flow** on first open:

 **Compulsory Permission Priming Pattern:** When any user opens the app, a glassmorphic modal prompts for hardware GPS access. Upon clicking *"Allow Live GPS Location Now"*, the browser native GPS triggers, the Leaflet map smoothly pans to their real coordinates, plots a pulsating cyan marker, and auto-switches to the nearest Himachal town hub.

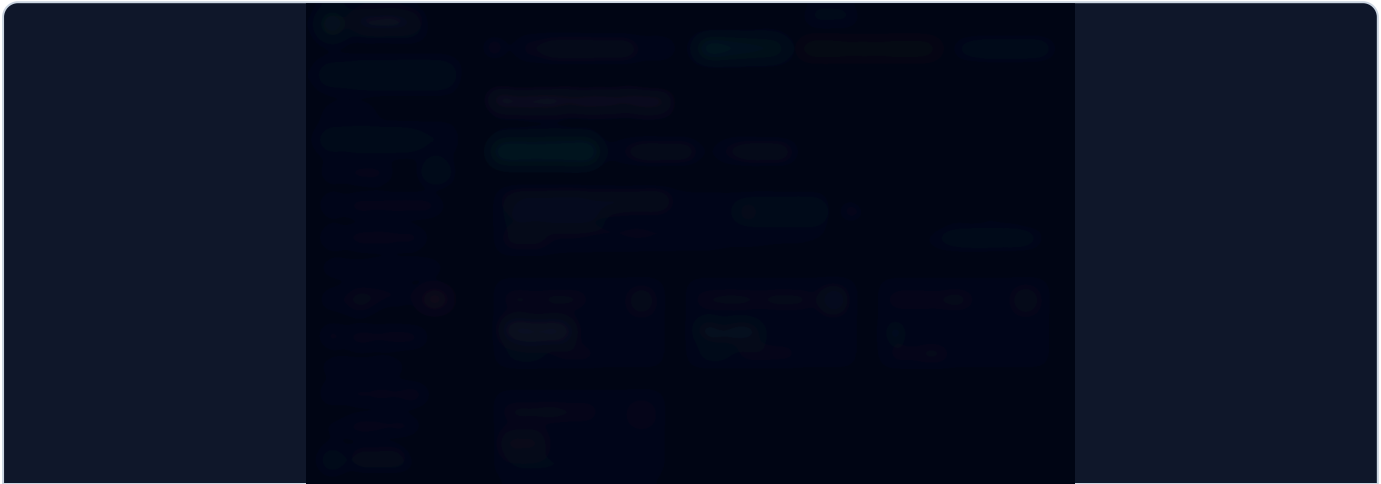


Figure 2.1: Universal Compulsory Location Prompt Modal on Initial Portal Launch

3. Real-Time Meteorological Weather Auto-Sync (Open-Meteo)

PahadiCart connects with the **Open-Meteo REST API** (100% free, zero billing, zero API keys) to dynamically sync real-time meteorological conditions for Himachal coordinates:

Weather Mode	WMO Weather Codes	Platform Protocol & Pricing Impact
☀️ Clear Skies	0, 1, 2, 3	Standard 45–60 min delivery SLA active across all mountain corridors. Base rates apply.
☁️ Monsoon Rain	51–67, 80–82, 95–99	+20 min buffer added to customer ETA; +₹15 Dynamic Rain Surge credited to rider.
❄️ Snow / Blizzard	71–77, 85–86	Safety Lock Active: Steep unpaved slopes locked. Pedestrian walking runners only.



Figure 3.1: Admin Command Tower with Live GPS Marker & Real-Time Open-Meteo Telemetry Badge

4. Universal Google Maps Navigation Deep Links

To connect the web application directly with native turn-by-turn hill GPS, PahadiCart generates standard universal Google Maps intent links: <https://www.google.com/maps/dir/?api=1&destination={lat},{lng}&travelmode=driving>

📖 1-Click Launch Across Portals:

- **Riders:** 1-click *"Open Turn-by-Turn in Google Maps"* on active delivery missions and batched mountain trails.
- **Customers:** *"Open Destination in Google Maps"* in the Live Mountain GPS Radar tracking screen.
- **Admins:** Direct Google Maps inspection from the Live Orders queue.

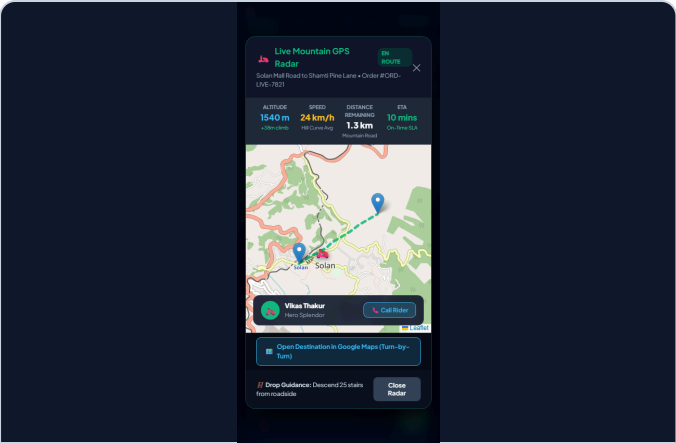


Figure 4.1: Customer Live Radar with Google Maps Action

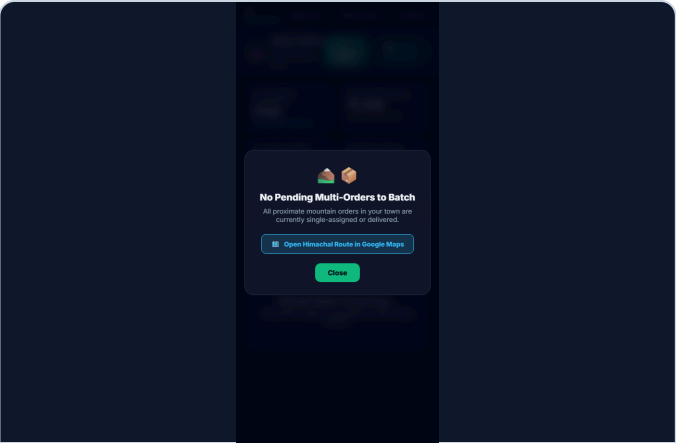


Figure 4.2: Rider Cockpit Batched Google Maps Navigation

5. End-to-End Mountain Order Triad Lifecycle

[Customer Order Placed] → Web Audio "Ghar Ki Ghanti" Chime | ▼ [Merchant KOT Slip] → 58mm Thermal Receipt Printed & Packed | ▼ [Rider Radar Ping] → Accepts & Climbs Batched Mountain Footpath | ▼ [Doorstep OTP Verified] → 4-Digit Secret Code Entered, Cash/UPI Collected | ▼ [Instant Settlement] → GMV, 1% TCS, Merchant Credit & Rider Payout Logged

6. Quick Troubleshooting & FAQ

Symptom	Root Cause	Resolution
GPS Prompt Not Appearing?	Session already confirmed in current tab.	Clear sessionStorage or open in Incognito to re-test the compulsory prompt modal.
Weather Sync Shows Default 18.2°C?	Network failure contacting Open-Meteo.	App automatically falls back to offline cache; click "Sync Live Weather" to retry.
Google Maps Button Does Not Open App?	Desktop browser without native app.	Universal URL seamlessly opens Google Maps Web in desktop browser with turn-by-turn routes.

PAHADICART • COMPLETE MASTER OPERATIONS BLUEPRINT & USER HANDBOOK
Engineered for the Steep Hills, Ancient Bazaars, and Resilient Communities of Himachal Pradesh • Version 2.9 (2026)
Live Production: <https://pahadicart.vercel.app> • **GitHub:** [HACK-X01/pahadicart](#)